Machine Learning Model Compression and Optimization for Vehicle Fuel Economy Prediction

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Abstract

This study investigates machine learning and neural network approaches for vehicle fuel economy prediction using two datasets: the Auto MPG dataset and the Vehicles Fuel Economy dataset. Linear Regression, Support Vector Regression (SVR), Random Forest Regression, and Multilayer Perceptron (MLP) models were implemented and evaluated. Model compression techniques including Principal Component Analysis (PCA), coreset selection, pruning, and quantization were applied to reduce model complexity while preserving predictive performance. Experimental results indicate that Random Forest consistently achieved the highest predictive accuracy across both datasets. Furthermore, dimensionality reduction significantly improved neural network performance on the high-dimensional Vehicles dataset.

1 Introduction

Fuel economy prediction is an important problem in transportation and environmental sustainability. Accurate predictive models assist manufacturers, policymakers, and consumers in understanding vehicle efficiency and fuel consumption.

This project evaluates multiple machine learning algorithms and compression techniques for predicting vehicle fuel economy. The objective is to compare predictive performance and investigate whether model complexity can be reduced without significantly degrading accuracy.

2 Datasets

2.1 Auto MPG Dataset

The Auto MPG dataset contains 398 vehicles and 9 attributes. The target variable is Miles Per Gallon (MPG). Features include cylinders, displacement, horsepower, weight, acceleration, model year, and origin.

The dataset is relatively small, low-dimensional, and contains only six missing horsepower values, making it a clean regression problem.

2.2 Vehicles Fuel Economy Dataset

The Vehicles dataset contains 47,702 vehicle records and 84 attributes. The target variable used in this study is `comb08`, representing combined fuel economy.

The dataset contains numerical, categorical, and boolean attributes, as well as substantial missing values in several columns. Data preprocessing included missing value handling, categorical encoding, and feature scaling.

Preprint.

3 Methodology

Four predictive models were evaluated:

1. Linear Regression
2. Support Vector Regression (SVR)
3. Random Forest Regression
4. Multilayer Perceptron (MLP)

Model compression and optimization techniques included:

- Principal Component Analysis (PCA)
- Coreset Selection
- Weight Pruning
- Quantization

Performance was evaluated using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the coefficient of determination (R^2).

4 Results

4.1 Auto MPG Dataset

Table 1: Auto MPG Dataset Results

Model	MSE	R^2
Linear Regression	8.196	0.848
SVR	6.814	0.873
MLP	5.122	0.905
Random Forest	**4.673**	**0.913**
PCA + MLP	6.150	0.886
Coreset MLP	7.268	0.865
Pruned MLP	5.166	0.904

Random Forest achieved the highest predictive performance on the Auto MPG dataset, achieving an R^2 score of 0.913. Pruning produced only a negligible reduction in accuracy, indicating that many neural network parameters were redundant. PCA slightly reduced performance because the dataset already contains a small number of informative features.

4.2 Vehicles Dataset

Table 2: Vehicles Dataset Results

Model	MSE	R^2
Linear Regression	97.963	0.200
SVR	121.856	0.005
MLP	116.083	0.052
Random Forest	**15.111**	**0.877**
PCA + MLP	16.786	0.863
Coreset MLP	108.010	0.118
Pruned MLP	118.766	0.031

The Vehicles dataset produced significantly different behavior. Random Forest substantially outperformed all other methods with an R^2 score of 0.877. The original MLP struggled on the high-dimensional dataset, achieving an R^2 score of only 0.052. However, PCA improved MLP performance dramatically, increasing the R^2 score to 0.863. This indicates that dimensionality reduction successfully removed redundant and noisy features.

5 Figures

The following figures were generated during experimentation: 1. Correlation Heatmap (Auto MPG Dataset)

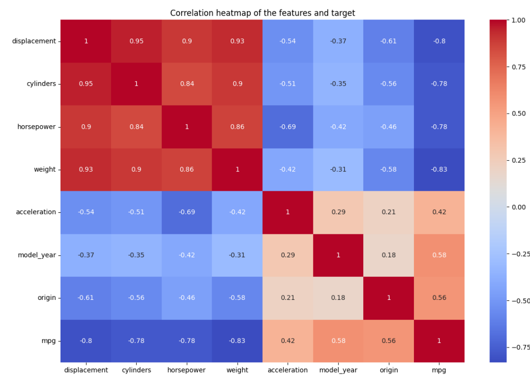

Figure 1: Enter Caption

2. Correlation Heatmap (Vehicles Dataset)

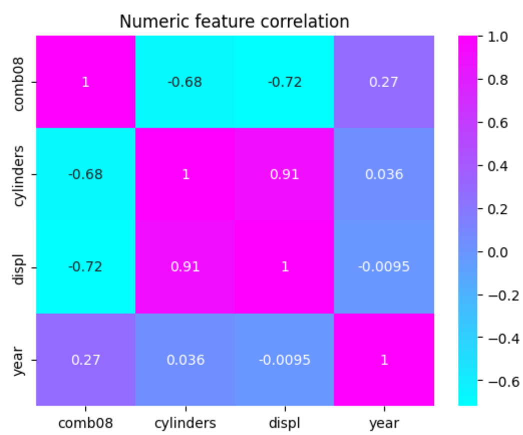

Figure 2: Enter Caption

Error bars were not included because each model was evaluated using a single train-test split rather than repeated cross-validation experiments.

6 Discussion

The experimental results demonstrate that Random Forest was the most robust model across both datasets. The Auto MPG dataset contained only 398 observations and 9 attributes, making it a relatively simple regression problem. In contrast, the Vehicles dataset contained 47,702 observations and 84 attributes, introducing substantial complexity.

Compression techniques had different effects depending on dataset characteristics. Pruning was highly effective on the Auto MPG dataset because the neural network contained redundant parameters.

Conversely, PCA proved most beneficial on the Vehicles dataset by reducing dimensionality and improving neural network learning.

7 Conclusion

This study compared machine learning and neural network approaches for vehicle fuel economy prediction. Random Forest consistently achieved the highest predictive accuracy on both datasets. Compression techniques demonstrated that model complexity can often be reduced with minimal performance degradation. In particular, PCA significantly improved neural network performance on high-dimensional vehicle data, highlighting the importance of dimensionality reduction in complex predictive modeling tasks.